DEPARTMENT OF

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

TITLE : Face Recognition Based Attendance Management System Using Deep Learning

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Content :

- INTRODUCTION
- PROBLEM DEFINITION
- PROPOSED SOLUTION
- SYSTEM REQUIREMENTS
- EXISTING SYSTEM
- PROPOSED SYSTEM
- MODULES
- ALGORITHM
- UML DIAGRAMS
- USE CASEDIAGRAM
- DATA FLOW DIAGRAM
- CONCLUSION

Introduction

Face-recognitionbasedattendance management system uses biometricauthenticationtoidentify individuals based on faces. The title is justified as it describes an automated system for tracking attendance using computer vision and deep learning methods, which offers improved accuracy, real-time monitoring, and enhanced security compared . Attendance management is a fundamental process in educational institutions.

Traditional methods such as manual roll calling and register- based systems are inefficient and susceptible to proxy attendance. To address these limitations, automated attendance systems using biometric technologies have been introduced. Among various biometric approaches, face recognition is a non-intrusive and effective technique. Recent advancements in deep learning have significantly improved the accuracy of face detection and recognition. By applying deep learning–based face recognition techniques, attendance can be recorded automatically in real time, ensuring accuracy, efficiency, and reliability

Problem Definition

Manual Process Inefficiency

Current manual attendance systems require significant time and effort to maintain, creating administrative burden for educational institutions.

Proxy Attendance Risk

Traditional methods allow possibility of proxy attendance, compromising the integrity of attendance records and institutional accountability.

Record Maintenance Challenges

Paper-based orfile-based attendance tracking systems are prone to errors, loss of data, and difficulty in generating comprehensive reports.

Proposed Solution

Oursystemimplementsan automatedface-recognition-based attendance management solution using the Multitask Convolutional Neural Network(MTCNN)algorithm forsuperioraccuracyand reliability.

01

ImageCapture

Students' images are captured in class to construct a comprehensive dataset for training and recognition.

02

FaceDetection&Recognition

MTCNN algorithm performs both face detection and recognition with high accuracy across varying conditions.

03

Data Upload

Recognized faces trigger automatic attendance data upload to the system database for record keeping.

04

Performance Validation

System achieves 98.2% accuracy, outperforming Haar Cascade CNN and other existing methods.

System Requirements:

Software Requirements

- Python programming environment
- OpenCV library for image processing
- MTCNN algorithm implementation
- Insight Face tool for face embedding
- SoftMax classifier for training
- Database management system
- GUI framework for user interface

Hardware Requirements

- High Resolution webcam or camera
- Processing unit with GPU support
- Adequate storage for dataset (500 images per person)
- Well lit classroom environment
- Network connectivity for data upload

Existing Systems Comparison

Variousfacerecognitionmethodshavebeenexploredforattendancesystems,each with distinct advantages and limitations:

Viola Jones Detector

Accuracy: Good for frontal faces

Limitation: Limited performance with varying illumination, increased training time

LBPH + Haar Cascade

Accuracy: 96.88%

Limitation: Requires well-lit environment, performance drops in poor lighting

Eigen Face + SVM

Accuracy: 61% with 50 images

Limitation: Low efficiency with small datasets, requires large face count

KNN Method

Accuracy: 99.27%

Limitation: High computational complexity, slower processing speed

Proposed System Architecture

1

Data Collection

500imagesper person captured via webcam, automatically stored with names for training dataset

2

MTCNN Processing

Three-stage CNN: P-Netforproposals, R-Net for refinement, O-Net for output with facial landmarks

3

Face Embedding

InsightFace generates128-value matrix representing crucial facial features for recognition

4

Classification

SoftMax classifier trains on embeddings and predicts student identity with high accuracy

MTCNN Algorithm: Three-Stage Process

Stage 1: Proposal Network (P-Net)

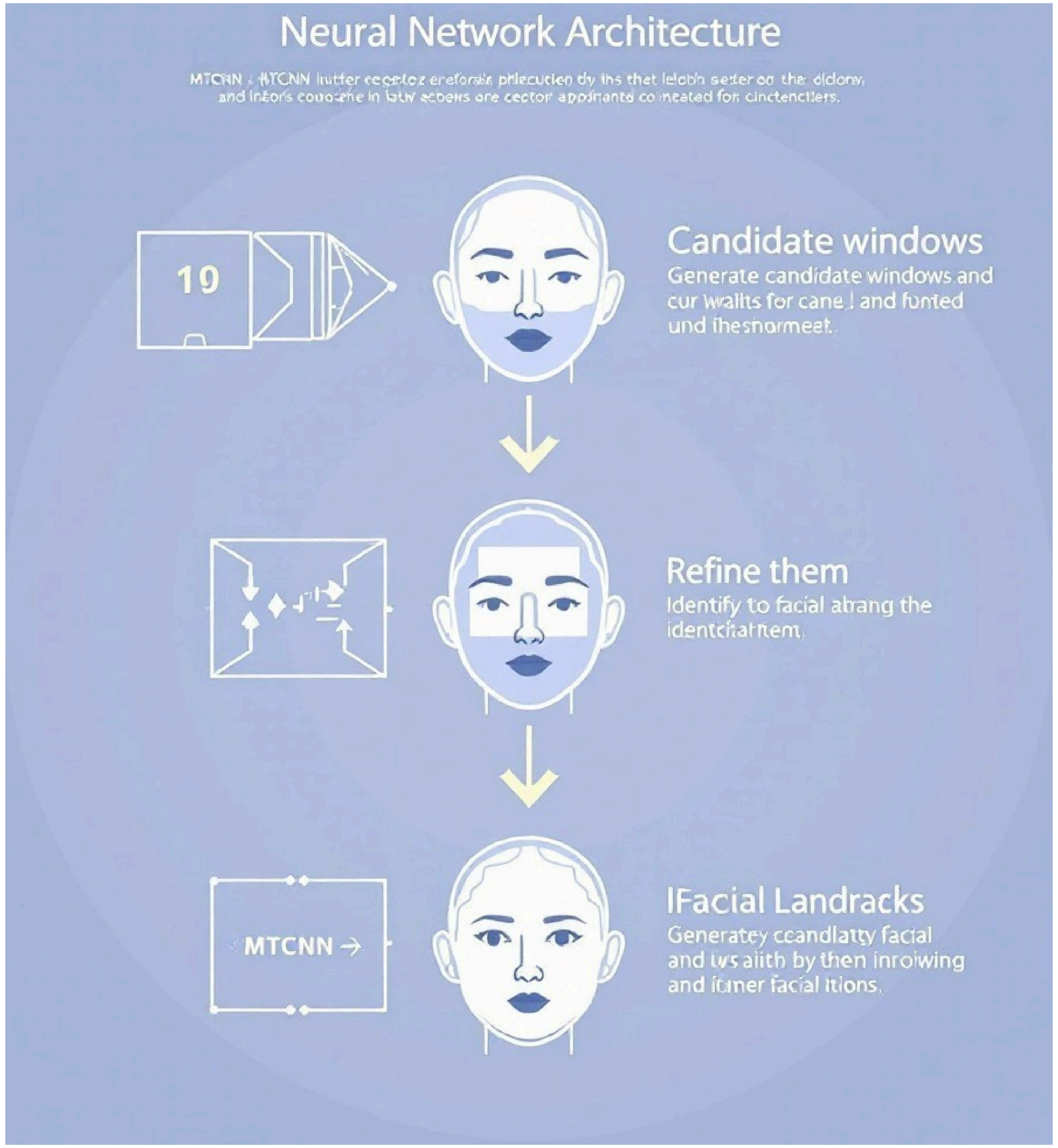
Generates candidatewindowsusingshallowCNN. Obtains bounding box regression vectors for face localization. Combines overlapping regions through refinement to reduce candidate count.

Stage 2: Refinement Network (R-Net)

Processes P-Netcandidatesthroughdeeperarchitecture. Uses bounding box regression for calibration and non-maximum suppression (NMS) for merging. Outputs 4-element bounding box vector and 10-element facial landmark vector.

Stage 3: Output Network (O-Net)

Provides detailedfaceanalysiswith precise landmark positions. Identifies five primary facial features: eyes, nose, mouth, and eyebrows. Produces final recognition result with high accuracy.



Modules

- Instead of focusing on human attendance, the system isnowadapted for face detection in educational and researches, involving students and faculty in fields like class room andExamhalls. The conventional “User” and “Admin” roles are replaced with “Student” and “Faculty” roles, and the MTCNN-based face recognition technology is repurposed for both secure user authentication and face feature detection.

1.User Registration Module :

- Purpose: Register new users into the system.
- Functions: Capture user face images.

Extract face encodings (feature vectors).

Save user data (ID, name, face encodings) into the database.

2.Face Detection Module :

- Purpose: Detect faces in real-time images or video frames.
- Functions:

Use MTCNN to locate faces in a frame.

Output bounding boxes and facial landmarks.

- Input: Image/frame from the camera.
- Output: Cropped face images for recognition

3.Face Recognition Module :

- Purpose: Identify the person in the detected face.
- Functions:Compare the detected face with database of registered faces.

Use deep learning model (e.g., FaceNet, ArcFace) for encoding and matching.

Output: User ID of recognized person.

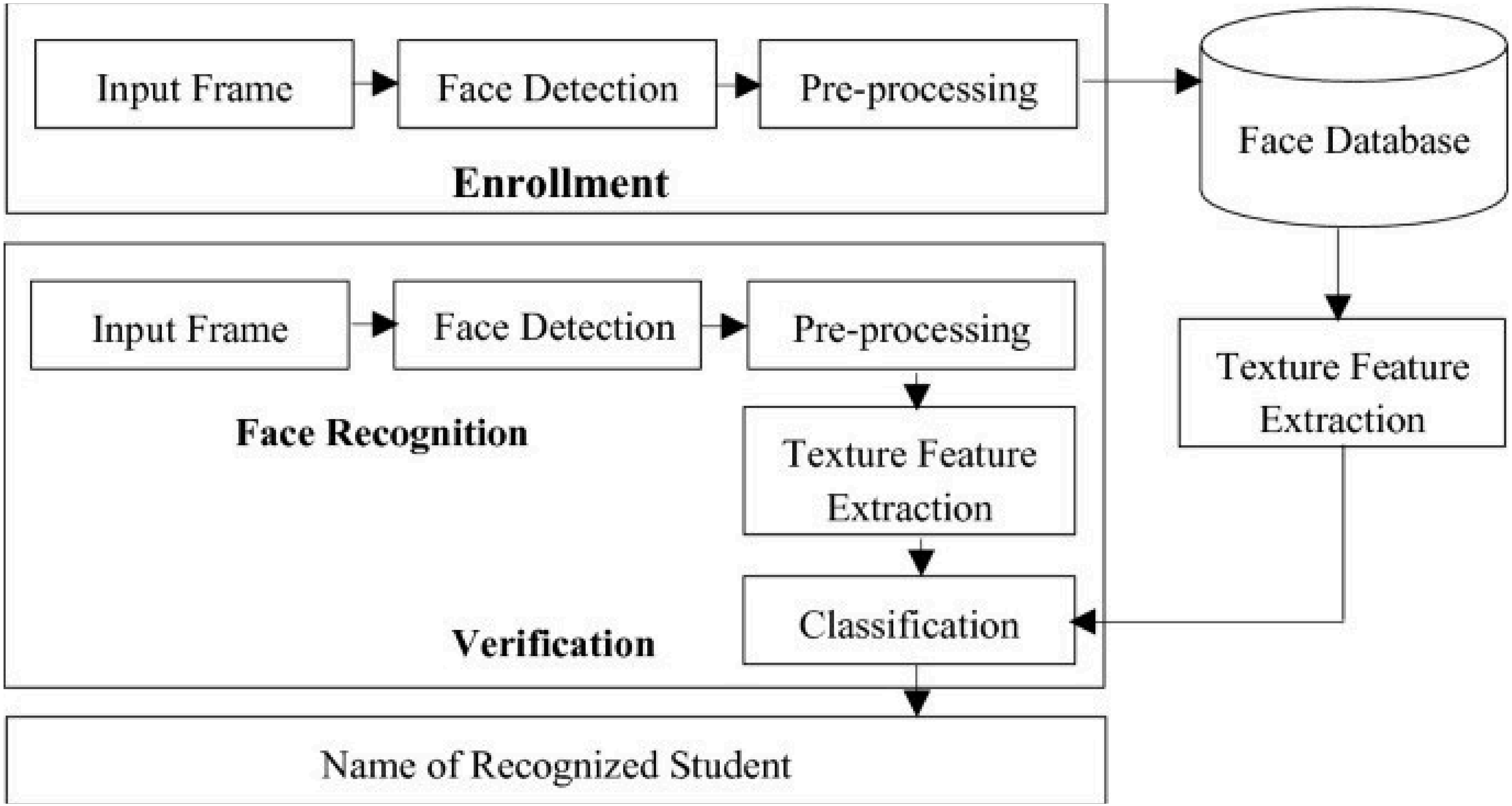
4.Attendance Logging Module :

- Purpose: Mark attendanc automatically.
- Functions:

Record user ID, timestamp, and status (Present/Absent).

Save the attendance record to the database.

SYSTEM ARCHITECTURE :



UML DIAGRAMS

UML Goal :

The Primary goals in the design of the UML are as follows:

1. Provideusersaready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development process.
4. Provide a formal basis for understanding the modeling language.
5. Support higher level development concepts such as collaborations, frameworks, patterns and components.
6. Integrate best practices.

Use Case Diagram

Record Students’ Profile & Information

Teacher registers and manages student details.

Detect Present Faces in Real-Time

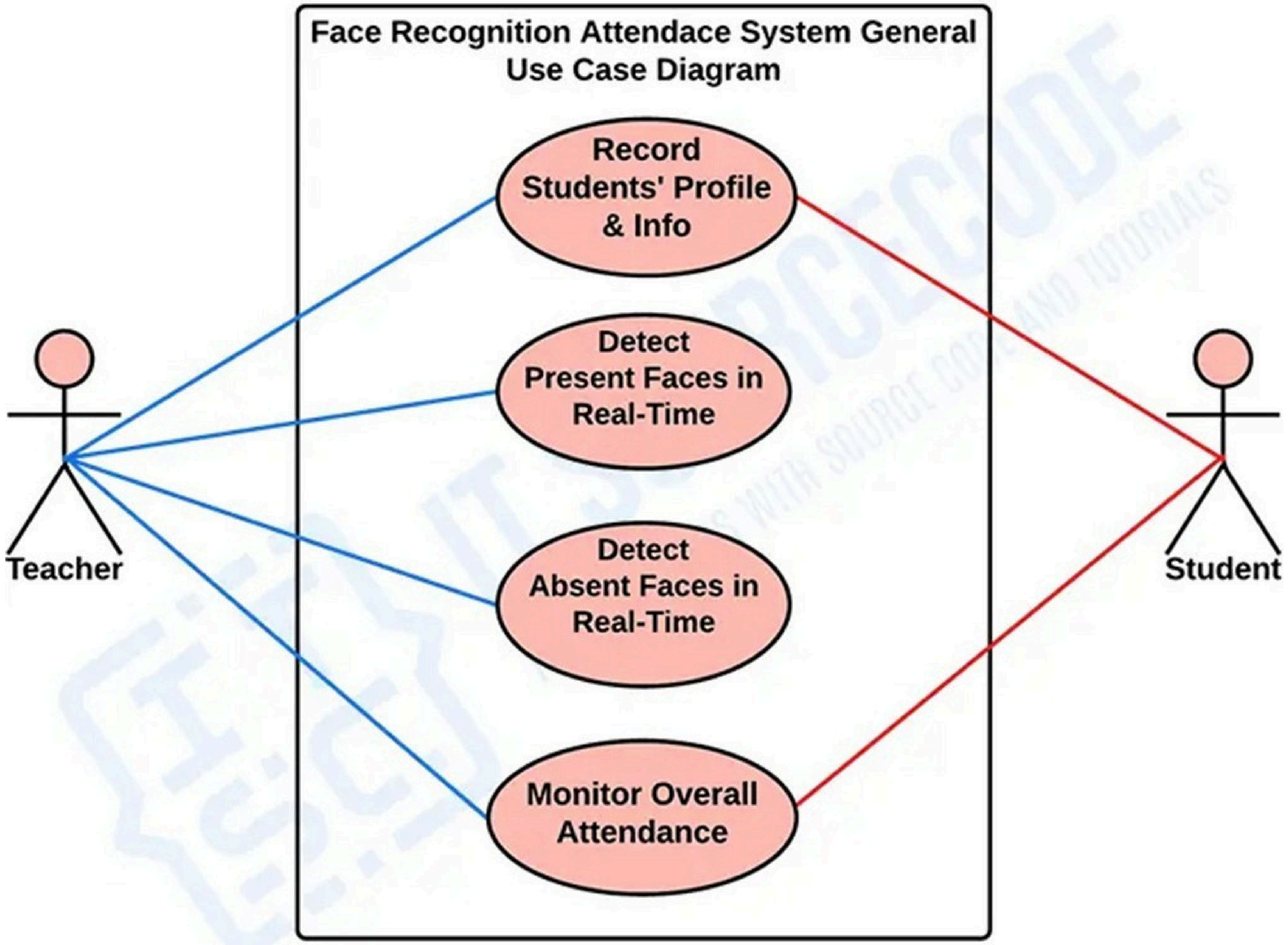
System detects students who are present.

Detect Absent Faces in Real-Time

System identifies students who are absent.

Monitor Overall Attendance

Teacher can view and analyze attendance records.



Class Diagram

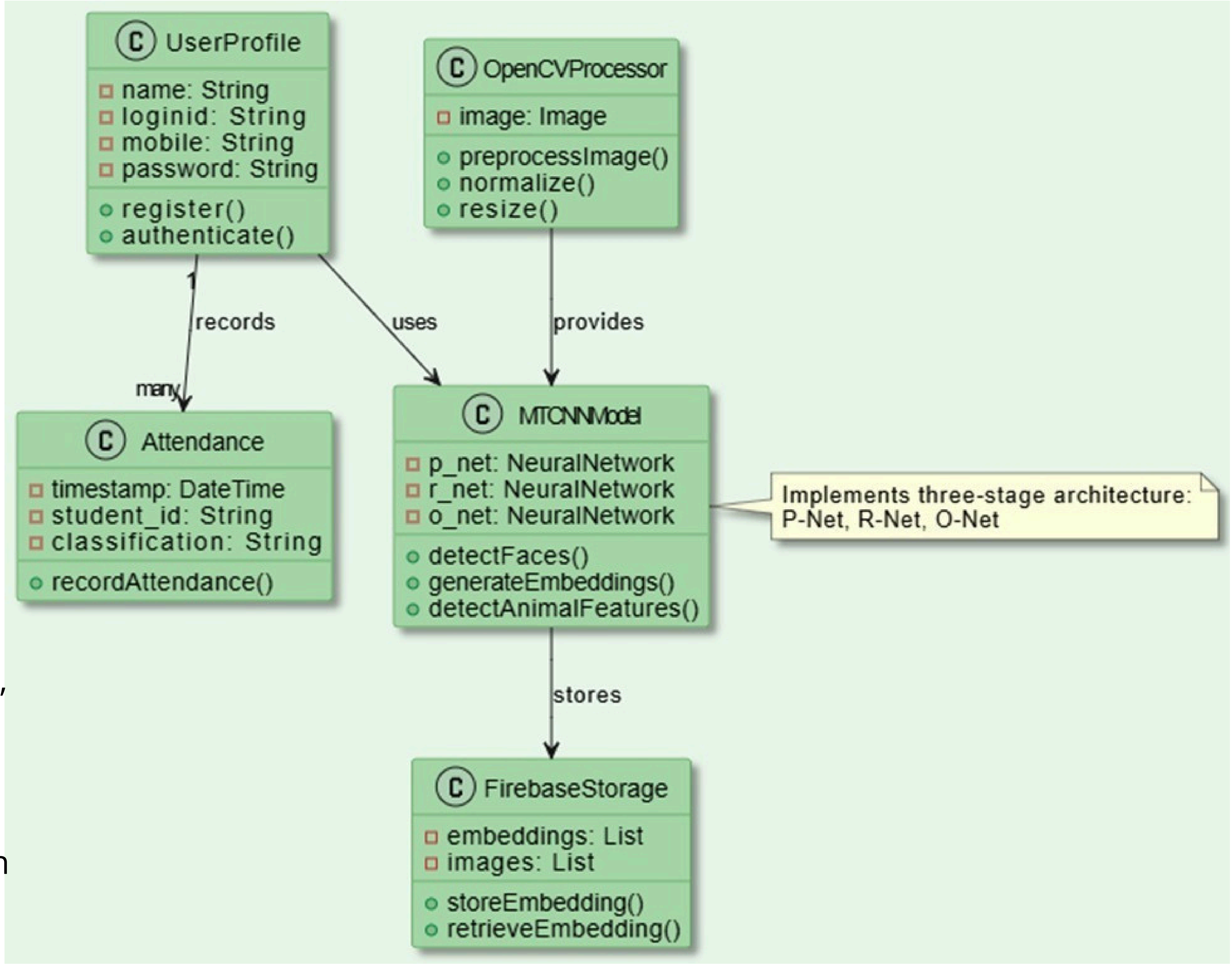
UserProfile :
Stores student details (name, login ID, mobile, password).
Handles registration and authentication.

Attendance :
Records attendance with timestamp, student ID, and classification.

OpenCVProcessor :
Preprocesses images using resize, normalize, and preprocessing.

MTCNNModel :
Detects faces and generates face embeddings using P-Net, R-Net, O-Net.

FirebaseStorage :
Stores face images and embeddings for future comparison



OBJECT DIAGRAM

Camera → Takes pictures of people.

FaceDetector (MTCNN) → Finds faces in the pictures.

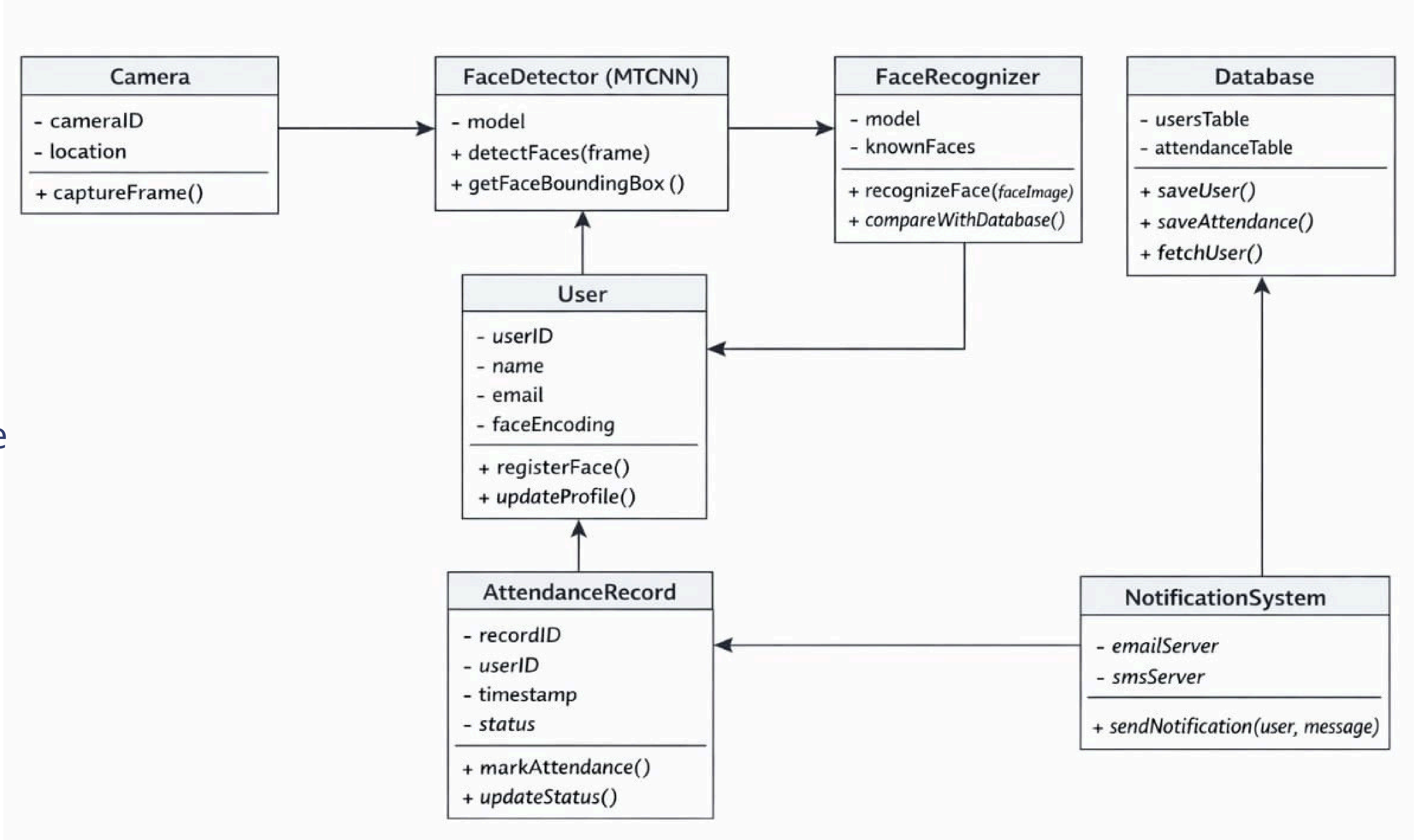
FaceRecognizer → Matches detected faces with registered users.

User → Represents each person with their ID, name, and face data.

AttendanceRecord → Marks and stores attendance for each user.

Database → Stores users and attendance records.

NotificationSystem → (Optional) Sends messages to users about their attendance.



SEQUENCE DIAGRAM

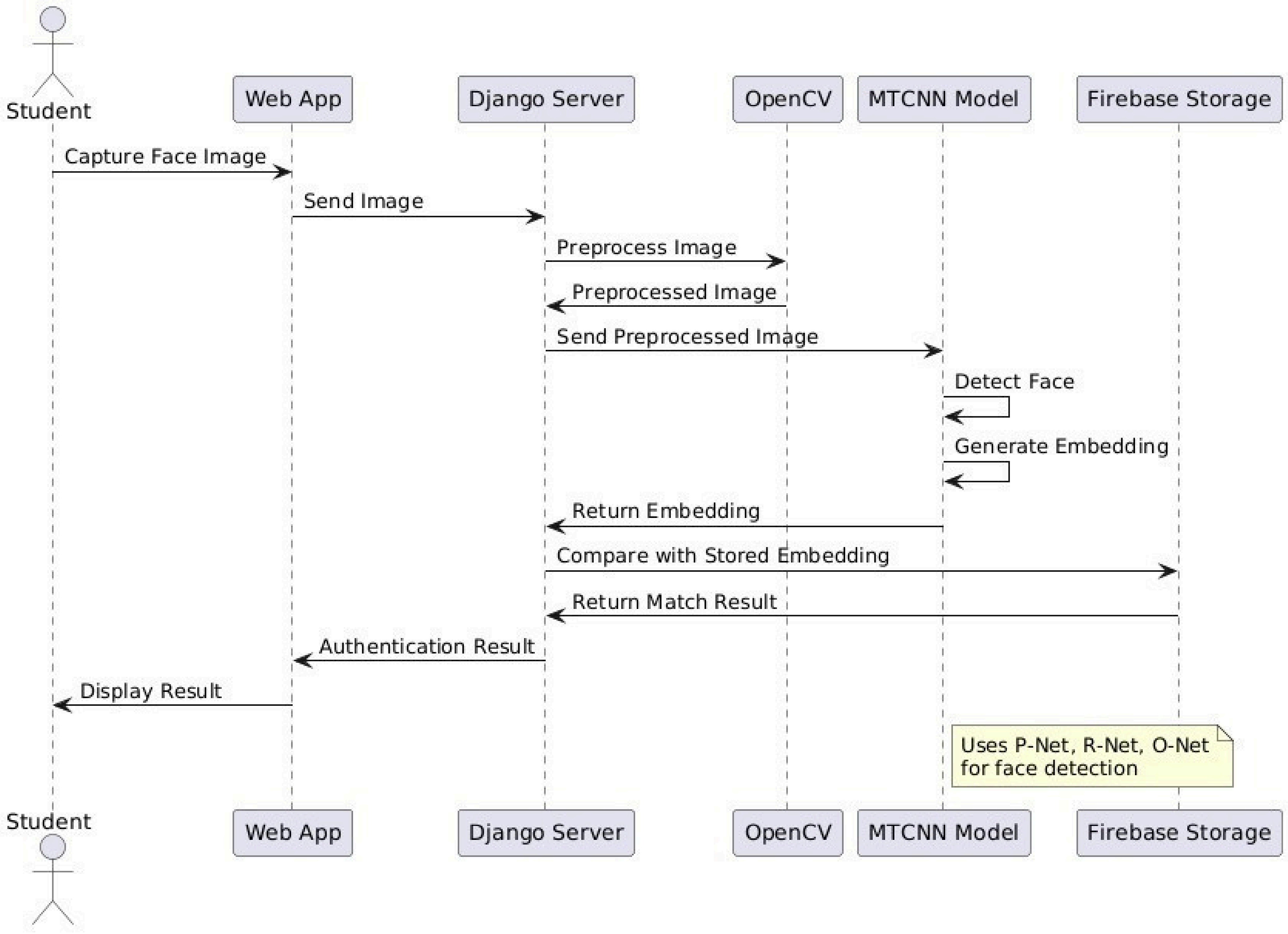
- Student captures face image using Web App.
- Image is sent to Django Server.
- Django sends image to OpenCV for preprocessing.
- Preprocessed image is sent to MTCNN Model.

MTCNN :

Detects face

Generates face embedding

- Embedding is compared with stored embeddings in Firebase.
- Match result is returned.
- Authentication result is shown in the web App.



ACTIVITY DIAGRAM

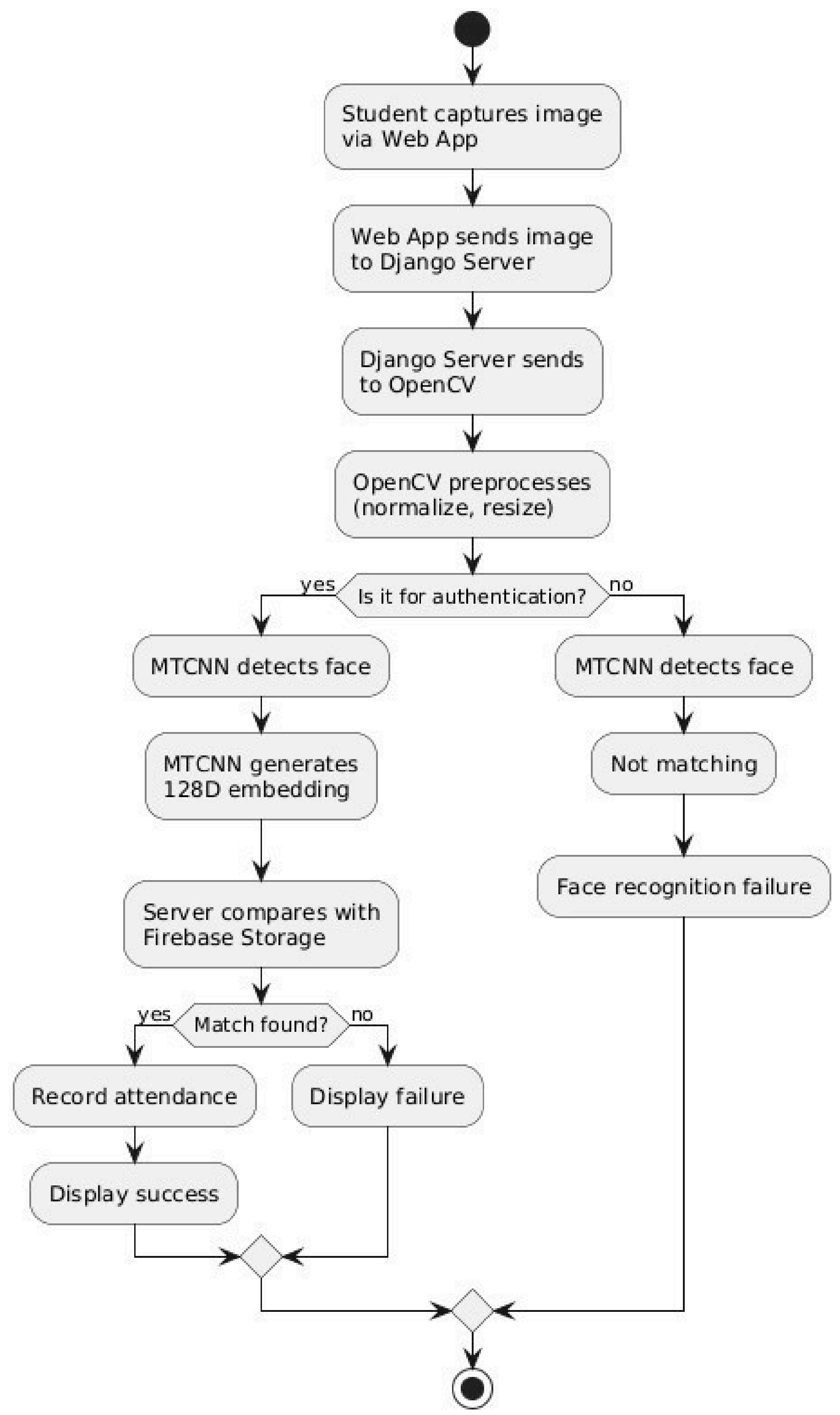
- Student captures image using web App.
- Image is sent to Django Server.
- OpenCV preprocesses the image.
- Decision:

  Authentication?

  Yes → MTCNN detects face → Generates 128D embedding

  Server compares with Firebase data
- If match found → Attendance recorded → Success shown

  If no match→ Failure shown
- Systemalsodisplays performance metrics like accuracy, precision, sensitivity.
-



- Web Application

Used by students for registration and attendance capture.

Sends face images to the backend using REST API.

- Face Recognition Pipeline

OpenCV Module – Preprocesses images (resize, enhancement).

MTCNN Detector – Detects faces from the image.

Face Embedding Model – Extracts unique facial features.

Softmax Classifier – Identifies the student.

Attendance Service – Marks attendance after recognition.

- Django Backend

Handles API requests, admin authentication, and attendance services.

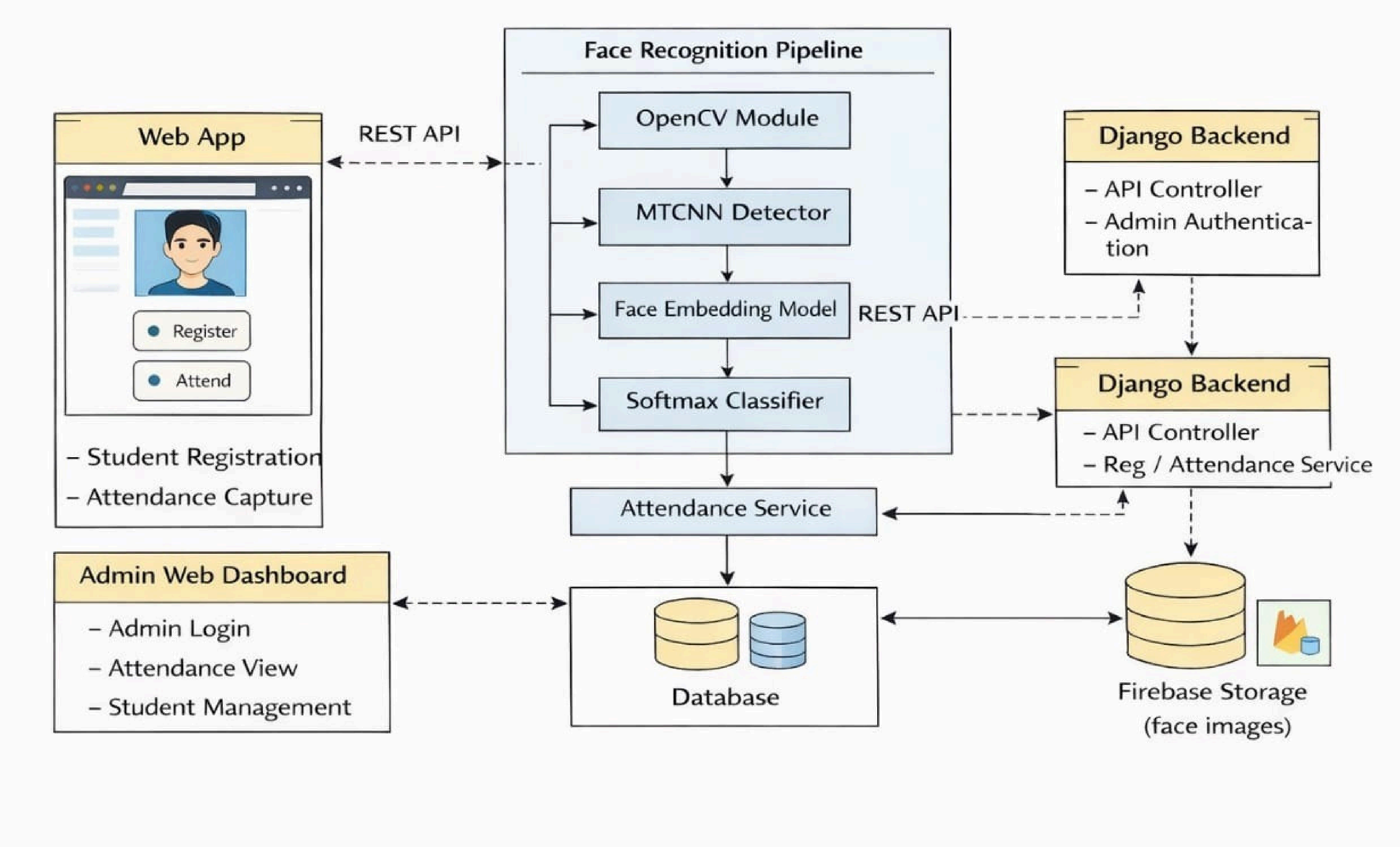
- Database

Stores student details and attendance records.

- Firebase Storage

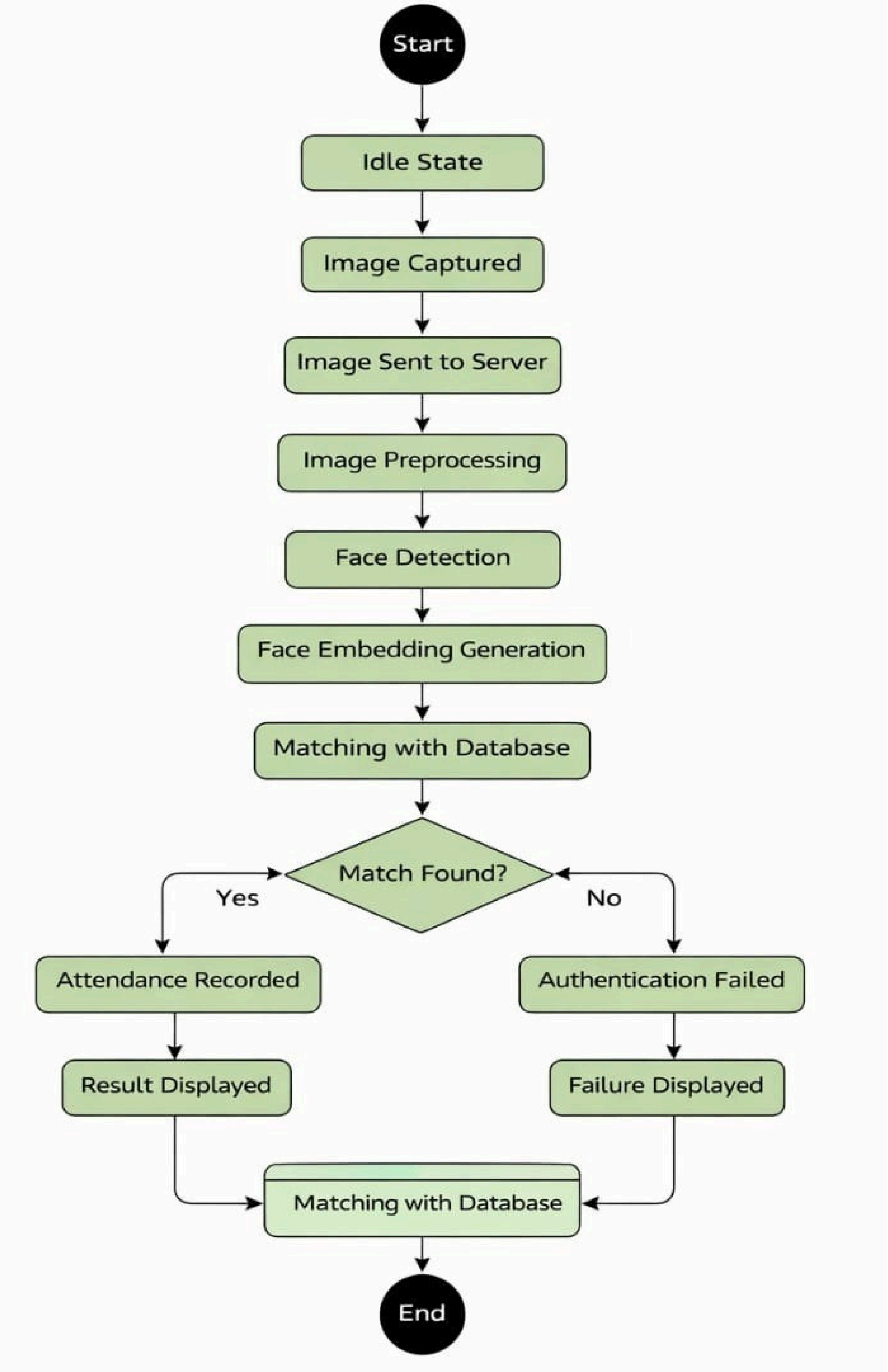
Stores face images used for training and recognition.

component Diagram



State Diagram

- The system starts in an Idle state, waiting for the student.
- The student captures a face image using the Android app.
- The image is sent to the server and preprocessed using OpenCV.
- The system detects the face and generates a face embedding using the MTCNN model.
- The generated embedding is matched with stored data in the database.
- If a match is found, attendance is recorded and a success message is shown.
- If no match is found, authentication fails and a failure message is displayed.



Deployment Diagram

- Student Mobile Device

Runs the Web App.
Captures images and sends them to the server via HTTPS.

- Django Server

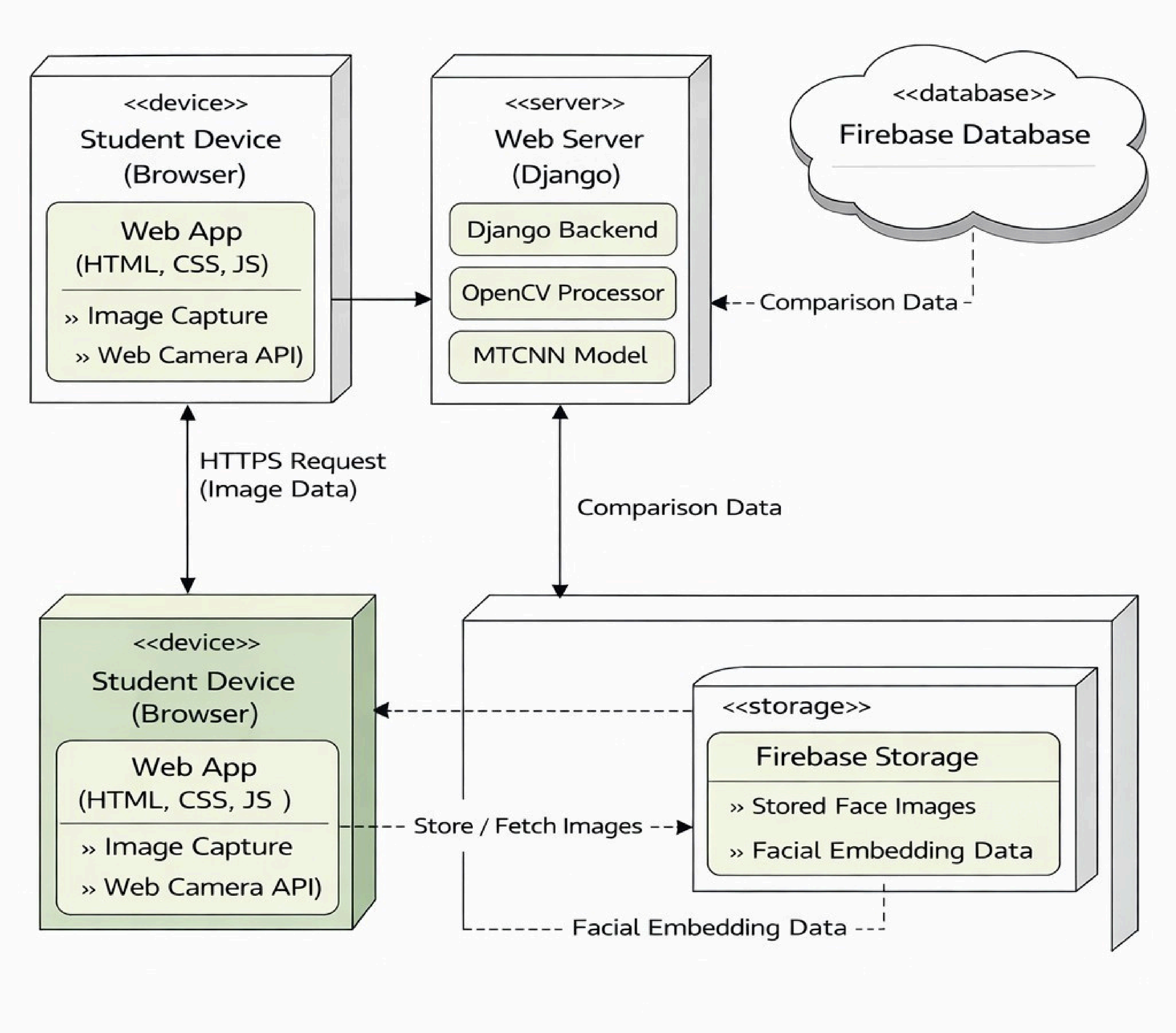
Hosts OpenCV processor and MTCNN model.
Performs face detection and recognition.

- Firebase Database

Stores comparison data, facial embeddings, and attendance records.

- Firebase Storage

Stores face images used during registration and recognition.



Package Diagram

- Web App : Used by students to capture face images.Sends images to the backend server.
- Backend / Django : Acts as the central controller. Receives images from the Web app

and sends them to facerecognition modules.

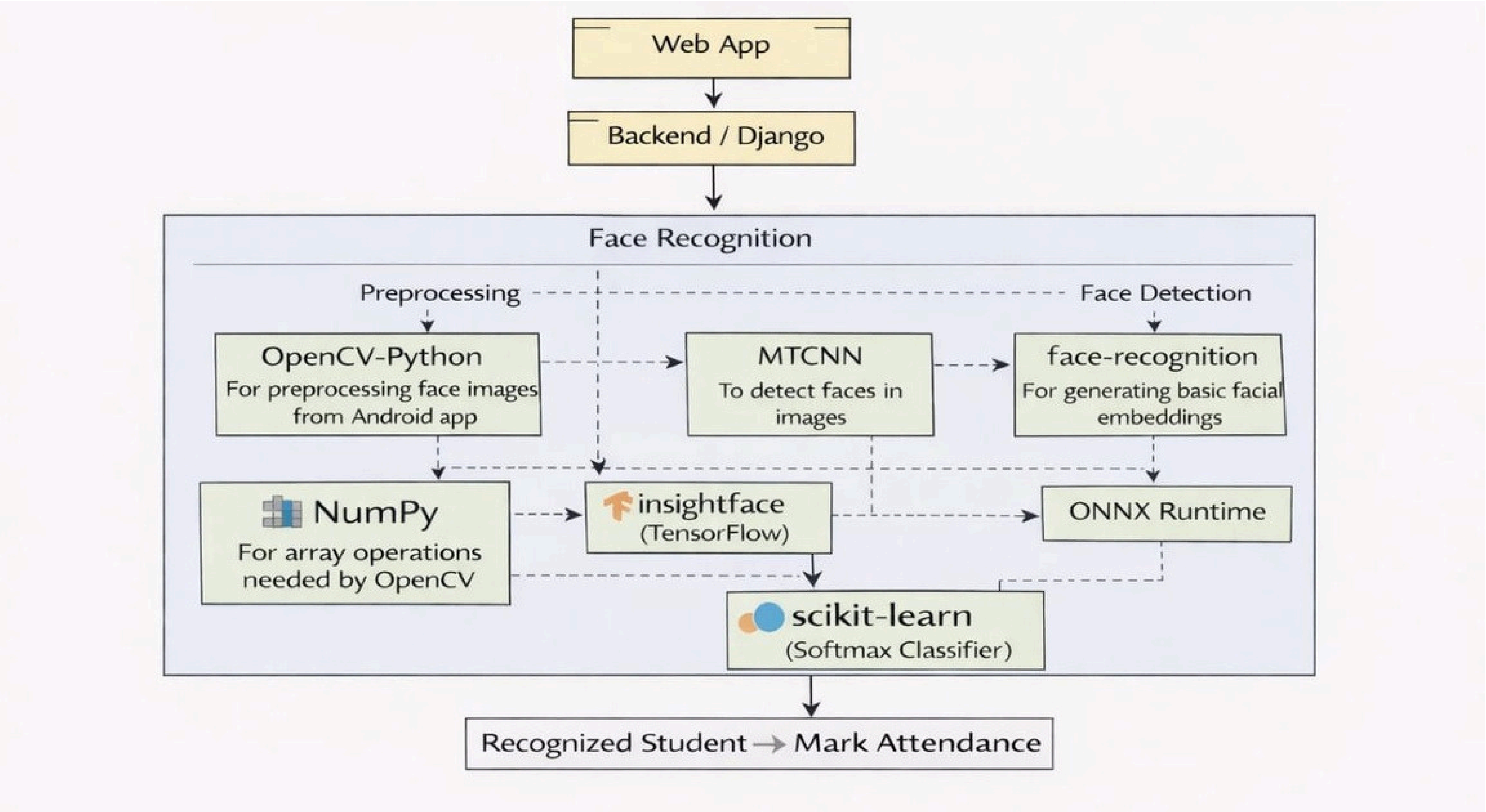
- OpenCV-Python : Performs image preprocessing like resizing and normalization.

- NumPy : Supports OpenCV by handling array and matrix operations.

- MTCNN : Detects faces from the input images.
- face-recognition / InsightFace (TensorFlow

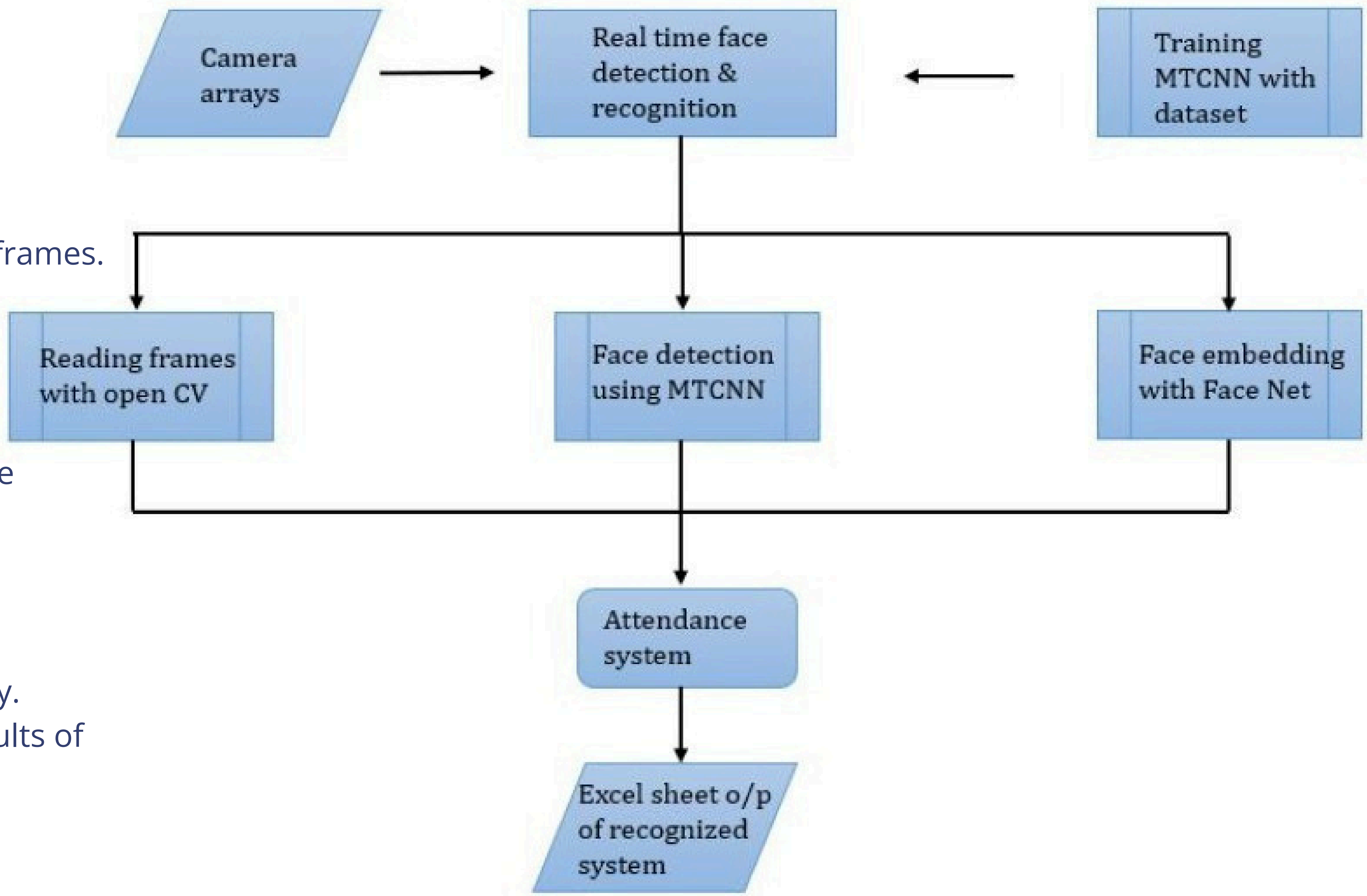
Extracts facial features (embeddings) from detected faces.

- ONNX Runtime : Helps in fast execution of deep learning models.
- scikit-learn (Softmax Classifier) : Classifies the face embeddings to identify the student.
- Final Output : If the student is recognized, attendance is marked



Data Flow Diagram

- Camera Arrays : Capture real-timeimages.
- OpenCV :Reads frames from the camera.
- MTCNN : Detects faces from the captured frames.
- FaceNet:: Generates face embeddings for recognition.
- Training MTCNN : Uses a dataset to improve accuracy.
- Attendance System
  Matches recognized faces.
  Marks attendance automatically.
- Excel Sheet Output : Stores attendance results of recognized students.



Conclusion

The proposed MTCNN-based system outperforms Haar Cascade CNN and other existing methods including LBP .Dual Variational Generation , and OpenFace . The three-stage cascaded architecture progressively refines face detection, achieving superior results even with head rotation, varying illumination, and multiple faces.

Future Enhancement: Integration of GSMmodule for automated SMS notifications to students and parents regarding attendance status.

Key Contribution: This automated system saves time, eliminates proxy attendance, reduces manual errors, and enhances institutional reputation through reliable, accurate attendance tracking.

Thank you